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BUILDINGS OF GENERALIZED BRAID GROUPS

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Introduction

The main result of this work is the following.

Theorem. Let V be a finite-dimensional real vector space, let $\mathcal{A}$ be a finite set of homogeneous hyperplanes of V, let $V_{\mathbb{C}}$ be the complexification of V, and put

$$Y = V_{\mathbb{C}} - \bigcup_{M \in \mathcal{A}} M_{\mathbb{C}}.$$

Assume that the connected components of

$$V - \bigcup_{M \in \mathcal{A}} M$$

are open simplicial cones. Then Y is a $K(\pi, 1)$.

Let V be as above, and let $W \subset \mathrm{GL}(V)$ be a finite group generated by reflections. Suppose that no nonzero vector of V is fixed by W:

$$V^W = 0.$$

Let φ be any Euclidean structure on V invariant under W, and let $\mathcal{A}$ be the set of hyperplanes M such that the orthogonal reflection with respect to M belongs to W. It is then known that $(V, \mathcal{A})$ satisfies the hypothesis of the theorem, and that W acts freely on the corresponding space Y_W. The quotient

$$X_W = Y_W / W$$

is therefore also a $K(\pi, 1)$.

This result had been conjectured by Brieskorn. It is new only for W of type H_3, H_4, E_6, E_7, or E_8 (Brieskorn [2]). We also give a new proof that the fundamental group of X_W is the generalized braid group corresponding to W (Brieskorn [3]).

Reduced to the special case considered above, the plan of the proof is as follows.

a) We construct a building $I(W)$ on which W acts, and a set $\mathcal{S}$ of spheres in $I(W)$, isomorphic to the unit sphere of V. The group W acts strictly simply transitively on $\mathcal{S}$.

b) We show that, up to homotopy, $I(W)$ is the bouquet of the spheres $S \in \mathcal{S}$.

c) We relate X_W and $I(W)$.

The description of $\mathcal{S}$ and the possibility of c) appeared during a conversation with Brieskorn and Tits in the spring of 1970. The ideas required to establish b) were supplied to me by Garside [4], whom I often follow very closely.

In Section 4 we determine the center of $\widetilde{W}$ and solve the word problem and the conjugacy problem in $\widetilde{W}$. These results were obtained independently by E. Brieskorn and K. Saito, who, like us, use Garside's methods [4].

The geometric language used and the techniques of proof are essentially due to Tits. I was able to become familiar with his ideas by attending one of his seminars and through conversations. I am pleased to express my gratitude to him here.

0. Notation

(0.1) $L^+(D)$: the free monoid with unit generated by a set D.

(0.2) $L(D)$: the free group generated by a set D.

(0.3) $\overline{A}$: the closure of a subset A of a topological space.

(0.4) For x, y in a monoid with unit and $n \geq 0$, put

$$\mathrm{prod}(n; x, y) = xyxy\cdots \quad (n \text{ factors});$$

thus

$$\mathrm{prod}(2n; x, y) = (xy)^n, \qquad \mathrm{prod}(2n+1; x, y) = (xy)^n x.$$

1. Galleries

(1.1) Let $\mathcal{A}$ be a finite set of hyperplanes, called walls, in a finite-dimensional real vector space V. We shall use the terminology of [1], V, §1: walls, chambers, faces, facets; the facets form a partition of V. The support P of a facet F is the intersection of the walls containing F; F is open in P. For every chamber A and every wall M, let $D_M(A)$ denote the set of chambers on the same side of M as A. If A and B are chambers, denote by $D(A, B)$ the intersection of the $D_M(A)$ for those M for which A and B are on the same side of M. Departing from the terminology of [1], IV, 1, ex. 15, we shall say that two chambers A and B are adjacent if $A \neq B$ and A and B have a face in common. We shall constantly use the following elementary facts.

(1.2) Lemma.

(i) Let M be a wall of a chamber B. There is a unique chamber B', adjacent to B, having M as a wall. The wall M is the only wall separating B from B'.

(ii) Let B_1, B_2, B_3 be three chambers, and let $\mathcal{M}(B_i, B_j)$ be the set of walls separating B_i from B_j. Then

$$\mathcal{M}(B_1, B_3) = \big(\mathcal{M}(B_1, B_2) - \mathcal{M}(B_2, B_3)\big) \cup \big(\mathcal{M}(B_2, B_3) - \mathcal{M}(B_1, B_2)\big).$$

A gallery of length n ($n \geq 0$) with source A and target B is a sequence of chambers $(C_0, \ldots, C_n)$ with $A = C_0$, C_{i+1} adjacent to C_i for $0 \leq i < n$, and $C_n = B$. The composite of two galleries $G = (C_0, \ldots, C_n)$ and $G' = (C'_0, \ldots, C'_m)$ is defined if $C_n = C'_0$ and is then

$$GG' = (C_0, \ldots, C_n, C'_1, \ldots, C'_m).$$

If G is a gallery with source A, then $A.G$ denotes its target.

The opposite gallery G^* of $G = (C_0, \ldots, C_n)$ is $(C_n, \ldots, C_0)$. One has $(GG')^* = G'^* G^*$. The antipodal gallery $-G$ is the sequence of antipodal chambers

$$-G = (-C_0, \ldots, -C_n).$$

One has $-(GG') = (-G)(-G')$ and $-(G^*) = (-G)^*$.

The distance $d(A, B)$ from a chamber A to a chamber B is the least length of a gallery from A to B. A gallery from A to B is minimal if its length is $d(A, B)$.

(1.3) Proposition. The distance $d(A, B)$ is the number of walls separating A from B. A gallery G from A to B is minimal if and only if it crosses once each wall which separates A from B, and crosses no other wall.

It is clear that every gallery from A to B crosses every wall separating A from B. It remains only to prove the existence of a gallery G from A to B of length equal to the number k of walls separating them. We argue by induction on k. The intersection of the $D_M(A)$ for M a wall of A is reduced to A. Hence either $A = B$, in which case one takes $G = (A)$, or there is a wall M of A separating A from B. Let A' be the chamber adjacent to A with wall M. The wall M is the only wall separating A from A', so a wall N separates A' from B if and only if $M \neq N$ and N separates A from B. By induction there is a gallery G' of length $k-1$ from A' to B, and we take $G = (A, A')G'$.

(1.4) Corollary. Let A, B, C be three chambers. The following conditions are equivalent:

(i) $d(A,C) = d(A,B) + d(B,C)$; equivalently, there exists a minimal gallery from A to C passing through B.

(ii) A wall separates A from C if and only if it separates A from B or B from C. In other words,

$$\mathcal{M}(A,C) = \mathcal{M}(A,B) \cup \mathcal{M}(B,C).$$

(iii) $B \in D(A,C)$.

(1.5) Proposition. Let P be an intersection of walls, let $V_P = V/P$, let $\mathrm{pr}_P : V \to V_P$ be the projection, and let $\mathcal{A}_P$ be the set of hyperplanes N of V_P such that $\mathrm{pr}_P^{-1}(N)$ is a wall. Denote by π'_P the unique map from facets of $(V, \mathcal{A})$ to facets of $(V_P, \mathcal{A}_P)$ such that $\mathrm{pr}_P(F) \subset \pi'_P(F)$.

(i) The map π'_P respects incidence $F_1 \subset \overline{F_2}$: if $F_1 \subset \overline{F_2}$, then

$$\mathrm{codim}(F_1 \text{ in } \overline{F_2}) \geq \mathrm{codim}(\pi'_P(F_1) \text{ in } \overline{\pi'_P(F_2)}).$$

It carries chambers to chambers: if A and B are adjacent, then either $\pi'_P(A) = \pi'_P(B)$ or $\pi'_P(A)$ and $\pi'_P(B)$ are adjacent.

(ii) Let F be a facet with support P. The restriction of π'_P to the set of facets E of $(V, \mathcal{A})$ such that $F \subset \overline{E}$ is bijective. Both this restriction and its inverse π_F preserve codimension, incidence $E_1 \subset \overline{E_2}$, and hence adjacency of chambers. We shall again write π_F for the inverse bijection between intersections of walls in V_P and intersections of walls in V which contain P.

(iii) If $C = \pi_F(C')$, then for every chamber X of $(V, \mathcal{A})$,

$$\pi'_P D(C, X) = D(C', \pi'_P(X)).$$

For $X = \pi_F(X')$, one has

$$\pi_F(D(C', X')) = D(C, X), \qquad \pi_F(\mathcal{M}(C', X')) = \mathcal{M}(C, X),$$

and π_F induces a bijection between minimal galleries from C' to X' and from C to X.

The verification is left to the reader; (iii) follows from 1.4(ii).

(1.6) Notation.

(i) If F is a facet with support P, then $V_P, \mathcal{A}_P, \mathrm{pr}_P, \pi_P$ will also be written $V_F, \mathcal{A}_F, \mathrm{pr}_F, \pi_F$.

(ii) Let P be an intersection of walls and C a chamber. Suppose that there is a facet of C, in $\overline{C}$, with support P. It is then unique; call it $F(P)$, and put

$$C.A(P) = \pi_{F(P)}(-\pi'_P(C)).$$

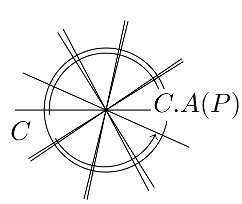
(1.7) Lemma.

(i) A wall separates C from $C.A(P)$ if and only if it contains P.

(ii) If M is a wall of C, then $C.A(M)$ is the unique chamber adjacent to C with wall M.

(iii) Let $M \neq N$ be two walls of C, and suppose that their intersection P contains a facet of C open in P. There are exactly two minimal galleries from C to $C.A(P)$. One begins with $(C, C.A(M))$, the other with $(C, C.A(N))$.

Assertion (iii) is checked by reduction to rank two, cf. 1.5(iii), where the picture is



(1.7.1)

From now on we assume the following condition.

(1.8) Hypothesis. The chambers are open simplicial cones. Equivalently, every chamber is the set of points with coordinates > 0 in a suitable basis of V.

(1.9.1) Remark. Let P be an intersection of walls.

(i) $(V_P, \mathcal{A}_P)$ still satisfies (1.8);

(ii) the set $\mathcal{A}^P$ of traces on P of the walls not containing P still satisfies (1.8).

If $\mathcal{A}$ is defined by a Weyl group W, see the introduction, then $\mathcal{A}^P$ is in general no longer of that type.

(1.9.2) Remark. Hypothesis (1.8) ensures that, if P is an intersection of walls of a chamber C, the condition of (1.6) is satisfied. The chamber $C.A(P)$ is therefore defined.

(1.10) Definition. Two galleries G and G' with the same endpoints are equivalent, written $G \sim G'$, if there exists a sequence of galleries $G = G_0, \ldots, G_n = G'$ $(n \geq 0)$ such that G_{j+1} is obtained from G_j as follows:

a) there are decompositions $G_j = E_1 F E_2$ and $G_{j+1} = E_1 F' E_2$;

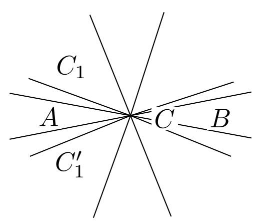
b) there is a chamber C and two walls M, N of C such that F and F' are the two minimal galleries from C to $C.A(M \cap N)$.

Composition of galleries, the operations $G \mapsto G^*$ and $G \mapsto -G$, the source and target maps, and the length function are compatible with this equivalence relation and therefore pass to the quotient. We shall generally denote equivalence classes of galleries by lowercase letters. We say that a class e begins, respectively ends, with a class f if there exists g with $e = fg$, respectively $e = gf$.

(1.11) Proposition. Equivalent galleries cross each wall the same number of times.

(1.12) Proposition. Two minimal galleries with the same endpoints are equivalent.

Let $G = (C_0, \ldots, C_n)$ and $G' = (C_0, C_1', \ldots, C_n')$ have endpoints A and B. We argue by induction on the length of the galleries. The case $n = 0$ is trivial, and the induction hypothesis permits us to suppose $C_1 \neq C_1'$. Let M and M' be the walls separating $A = C_0 = C_0'$ from C_1 and C_1', and put $C = A.A(M \cap M')$. The walls M and M' separate A from B. By reduction to rank two (1.5)(iii), every wall separating A from C also separates A from B.



(drawing in $V_{M\cap M'}$; we omit writing $\pi'_{M\cap M'}(\ )$).

Let F (respectively F') be the minimal gallery from A to C passing through C_1 (respectively through C_1'), and let E be a minimal gallery from C to B. The galleries FE and $F'E$ are minimal and equivalent. The minimal galleries G and FE, respectively G' and $F'E$, begin with (A, C_1), respectively with (A, C_1'). The induction hypothesis implies that they are equivalent, and (1.12) follows.

(1.13) Notation.

(i) Let $u(A, B)$ denote the equivalence class of minimal galleries from A to B.

(ii) If I is a set of walls of a chamber C, with intersection P, put

$$A(P) = u(C, C.A(P))$$

(cf. (1.9.2)). If I is the set of all walls, put $A = u(C, -C)$.

The interest of the notation lies in its ambiguity with respect to C.

(1.14) Proposition. Let A be a chamber and let $\mathfrak{a}$ be a set of gallery classes of bounded length and source A. Suppose that $\mathfrak{a}$ satisfies condition (i):

(ia) $(A) \in \mathfrak{a}$;

(ib) if $gh \in \mathfrak{a}$, then $g \in \mathfrak{a}$;

(ic) let g have target B, and let M, N be two walls of B. If $gA(M)$ and $gA(N)$ are in $\mathfrak{a}$, then $gA(M \cap N)$ is in $\mathfrak{a}$.

Then:

(ii) there is a unique gallery class x with source A such that

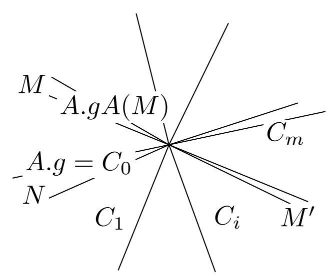
$$\mathfrak{a} = \{g \mid x \text{ begins with } g\}.$$

Uniqueness is clear: if x begins with y and $x \neq y$, then y has strictly smaller length than x. Let x be of maximal length in $\mathfrak{a}$. To see that it works, it is enough to prove the following assertion:

(∗) Let g and M be such that (a) x begins with g, (b) x does not begin with $gA(M)$, and $gA(M) \in \mathfrak{a}$. Then there exist g' and M' satisfying again (a) and (b), with g' strictly longer than g.

Since $gA(M) \in \mathfrak{a}$, maximality of x implies $g \neq x$, hence x begins with $gA(N)$ for a suitable N, necessarily distinct from M. By (ic), $gA(M \cap N) \in \mathfrak{a}$.

Since $gA(M \cap N)$ begins with $gA(M)$, x does not begin with $gA(M \cap N)$. Let $(C_0, \ldots, C_m)$ be the minimal gallery from $A.g$ to $A.gA(M \cap N)$ which begins with $(A.g, A.gA(N))$. Let i $(0 < i < m)$ be the largest index such that x begins with $g' = g(C_0, \ldots, C_i)$. Then g' and the wall M' between C_i and C_{i+1} satisfy (a)(b).



(projection into $V_{M \cap N}$).

(1.15) Proposition. Let A be a chamber and let $\mathfrak{B}$ be a set of chambers. For $\mathfrak{B}$ to satisfy condition (i):

(ia) $A \in \mathfrak{B}$;

(ib) if $B \in \mathfrak{B}$, then $D(A, B) \subset \mathfrak{B}$;

(ic) let M, N be two walls of a chamber B. If A and B are on the same side of M and of N, and $B.A(M)$ and $B.A(N)$ belong to $\mathfrak{B}$, then $B.A(M \cap N) \in \mathfrak{B}$,

it is necessary and sufficient that:

(ii) there exists a unique chamber C such that $\mathfrak{B} = D(A, C)$.

It is easy to check that (ii) implies (i). To prove the converse, apply (1.14) to the set $\mathfrak{a}$ of the $u(A, B)$ for $B \in \mathfrak{B}$, and then apply (1.4)(i) to the gallery class $x = u(A, X)$ obtained.

(1.16) Corollary. Let A and B be adjacent chambers separated by the wall M, and let C be on the same side of M as B. There exists C' such that

$$D(A, C) \cap D_M(B) = D(B, C').$$

(1.17) Corollary.

(i) Let A, C_1, C_2 be three chambers. There exists C such that

$$D(A, C) = D(A, C_1) \cap D(A, C_2). \tag{1.17.1}$$

(ii) Let A and C_1 be two chambers, and let M be a wall of A. There exists C' such that

$$D(A, C') = D(A, C_1) \cap D_M(A). \tag{1.17.2}$$

Indeed, (ii) is the special case of (i) with $C_2 = -(A.A(M))$.

(1.18) Lemma. Let M, M', M'' be three distinct walls of a chamber A, and put

$$A_1 = A.A(M), \qquad B = A.A(M \cap M' \cap M''),$$

$$B' = A.A(M \cap M'), \qquad B'' = A.A(M \cap M'').$$

For any chamber C, if $B' \in D(A_1, C)$ and $B'' \in D(A_1, C)$, then $B \in D(A_1, C)$.

The facet F of A open in $M \cap M' \cap M''$ is a facet of all the chambers A, A_1, B, B' and B''. By (1.5)(iii), we may reduce to the case $\dim V = 3$. By (1.17)(ii), we may also suppose that A_1 and C are on the same side of M. Under these hypotheses, if $B', B'' \in D(A_1, C)$, then M' and M'' separate A_1 from C, while A_1 and C remain on the same side of M. Thus A and C are separated by M, M' and M'', and $C = -A = B$.

The following key result is inspired by [4].

(1.19) Proposition.

(i) Let A, B, C be three chambers, F a gallery from A to B, and G_1, G_2 two galleries from B to C. If $FG_1 \sim FG_2$, then $G_1 \sim G_2$.

(ii) Similarly, if $G_1E \sim G_2E$, then $G_1 \sim G_2$.

(iii) Let g be an equivalence class of galleries with source A. There exists a chamber C such that g begins with $u(A, B)$ if and only if $B \in D(A, C)$.

Statement (ii) follows from (i) by passing to opposite galleries.

Let $G = (C_0, \ldots, C_n)$ and $G' = (C'_0, \ldots, C'_n)$ be two galleries of length n with the same endpoints. If $n \geq 1$, set $G_1 = (C_1, \ldots, C_n)$ and $G'_1 = (C'_1, \ldots, C'_n)$. If in addition $C_1 \neq C'_1$, let M and M' be the walls separating $C_0 = C'_0$ from C_1 and C'_1, and put $A = C_0.A(M \cap M')$.

Consider the following relation $R_n(G, G')$. It means that one of the following holds:

a) $n = 0$;

b) $n \neq 0$, $C_1 = C'_1$, and $G_1 \sim G'_1$;

c) $n \neq 0$, $C_1 \neq C'_1$, and there exists a gallery F from A to $C_n = C'_n$ such that

$$G_1 \sim u(C_1, A)F, \qquad G'_1 \sim u(C'_1, A)F.$$

We prove by induction on n that (A_n): R_n is an equivalence relation.

Assume (A_i) for $i < n$, and prove the three intermediate assertions below.

(1.19.1) For galleries of length $i < n$, $R_i(G, G')$ is equivalent to $G \sim G'$.

The implication $R_i(G, G') \Rightarrow G \sim G'$ is trivial, and if $G \sim G'$ then, using the notation of (1.10), one has $R_i(G_j, G_{j+1})$.

(1.19.2) Assertion (i) is true for composites FG_i of length $< n$. This follows from (1.19.1) and the second clause in the definition of the relations R_i.

(1.19.3) Assertion (iii) is true for g of length $< n$.

This follows from (1.15) applied to the set $\mathfrak{B}$ of chambers B such that g begins with $u(A, B)$. The hypothesis (ic) of (1.15) is checked using (1.19.2), (1.18), and the third clause in the definition of the R_i.

It remains to prove that if G, G' and G'' are three galleries of length n from A to C, with $R_n(G, G')$ and $R_n(G, G'')$, then $R_n(G', G'')$. If $n = 0$, or if $C_1 = C'_1$, or if $C_1 = C''_1$, this is trivial.

For $n > 0$, let M, M', M'' be the walls separating $A = C_0 = C'_0 = C''_0$ from C_1, C'_1, C''_1. There are two cases.

Case 1. $C_1 \neq C_1' = C_1''$. Let $B = A.A(M \cap M')$. By hypothesis,

$$(C_1, \dots, C_n) \sim u(C_1, B)F' \sim u(C_1, B)F'',$$

$$(C_1', \dots, C_n') \sim u(C_1', B)F', \qquad (C_1'', \dots, C_n'') \sim u(C_1'', B)F''.$$

By (1.19.2), $F' \sim F''$, hence $R_n(G', G'')$.

Case 2. C_1, C_1', C_1'' are pairwise distinct. Put

$$B_1 = A.A(M' \cap M''), \quad B' = A.A(M \cap M'), \quad B'' = A.A(M \cap M''),$$

and $B = A.A(M \cap M' \cap M'')$. Then

$$G_1 \sim u(C_1, B')F' \sim u(C_1, B'')F'',$$

$$G_1' \sim u(C_1', B')F', \qquad G_1'' \sim u(C_1'', B'')F''.$$

Thus the class g_1 of G_1 begins with $u(C_1, B')$ and $u(C_1, B'')$. By (1.19.3) and Lemma (1.18), g_1 begins with $u(C_1, B)$:

$$G_1 \sim u(C_1, B)F.$$

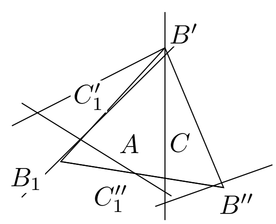
By (1.19.2), $F' \sim u(B', B)F$ and $F'' \sim u(B'', B)F$. Hence

$$G_1' \sim u(C_1', B)F, \qquad G_1'' \sim u(C_1'', B)F.$$

Therefore

$$G_1' \sim u(C_1', B_1)u(B_1, B)F, \qquad G_1'' \sim u(C_1'', B_1)u(B_1, B)F,$$

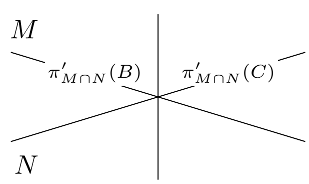
and so $R_n(G', G'')$.



(a drawing on the unit sphere of $V_{M \cap M' \cap M''}$).

(1.20) Corollary. The conditions (i) and (ii) of (1.14) are equivalent.

Assume (ii) and prove (ic). Under the hypotheses of (ic), if $x = gh$, with h of source B, then (1.19)(i) implies that h begins with $A(M)$ and $A(N)$. Let C be the chamber guaranteed by (1.19)(iii) for h. Since M and N separate B from C, $\pi'_{M\cap N}(C)$ can only be $\pi'_{M\cap N}(B).A$.



And (ic) follows from (1.5)(iii).

(1.21) Corollary. For every chamber A, let $n(A, i)$ be the number of gallery classes with source A and length i. Put

$$f_A = \sum_0^\infty n(A, i)t^i \in \mathbb{Z}[[t]].$$

For every chamber A, and for P running through the intersections of walls of A, including $\{0\}$ and V, one has

$$\sum_P (-1)^{\operatorname{codim} P} t^{d(A, A.A(P))} f_{A.A(P)} = 1.$$

This expresses the facts that:

a) the number of gallery classes of length i which begin with $A(P)$ is

$$n(A.A(P), i - d(A, A.A(P)));$$

b) gallery classes which begin with both $A(P)$ and $A(Q)$ begin with $A(P \cap Q)$.

(1.22) Algorithm. Let $G = (A_0, \ldots, A_n)$ be a gallery of length $n > 1$, let $G_1 = (A_1, \ldots, A_n)$, let M be the wall separating A_0 from A_1, let C be the chamber whose existence is guaranteed by (1.19)(iii) for G, and let C_1 be the analogous chamber for G_1. One can compute C recursively using the formula

$$D(A_1, C) = D_M(A_1) \cap D(A_1, C_1). \tag{1.22.1}$$

Since $A_1 \in D(A, C)$, the wall M separates A from C, and $C \in D_M(A_1)$. By (1.19)(i), one also has $D(A_1, C) \subset D(A_1, C_1)$, giving one inclusion. Finally, if $B \in D_M(A_1) \cap D(A_1, C_1)$, then G_1 begins with $u(A_1, B)$ and G begins with

$$u(A, A_1)u(A_1, B) = u(A, B),$$

which proves (1.22.1).

(1.23) Corollary. Let G and H be two composable galleries. Let A be the source of G, B the source of H, and let C be as in (1.19)(iii), so that

$$H \sim u(B, C)H'.$$

Then, for every chamber D, the class of GH begins with $u(A, D)$ if and only if the class of $Gu(B, C)$ begins with $u(A, D)$.

(1.24) Proposition. Let A be a chamber, let P be an intersection of walls of A, let $F = F(P)$ (1.6), and let g be an equivalence class of galleries with source A. There exists a gallery class g' of $(V_P, \mathcal{M}_P)$, with source $\pi_F^{-1}(A)$, such that, for every gallery H of $(V_P, \mathcal{M}_P)$, g begins with $\pi_F(H)$ if and only if g' begins with H.

Let $\mathcal{S}$ be the set of gallery classes h with source $\pi_F^{-1}(A)$ in $(V_P, \mathcal{M}_P)$ such that g begins with $\pi_F(h)$. Apply (1.14) to $(V_P, \mathcal{M}_P)$ and to $\mathcal{S}$. By (1.19), condition (i) is satisfied, cf. the proof of (1.20), and (1.24) follows from (1.14)(ii).

(1.25) The set of galleries may be regarded as the set of arrows of a category $\mathrm{Gal}_0(V, \mathcal{M})$ whose objects are the chambers. Similarly, equivalence classes of galleries are the arrows of a quotient category $\mathrm{Gal}^+(V, \mathcal{M})$. Let A, B, C be three chambers, let E be a gallery from A to B, and let F be a gallery from B to C. Although the general categorical convention is different, we shall continue to write EF for the composite of E and F. The law $G \mapsto G^*$, respectively $G \mapsto -G$, is an anti-equivalence, respectively an equivalence, of $\mathrm{Gal}_0(V, \mathcal{M})$, or of $\mathrm{Gal}^+(V, \mathcal{M})$, with itself, inducing the identity, respectively $C \mapsto -C$, on objects.

When no confusion is possible, we shall write simply Gal_0 and Gal^+ for $\mathrm{Gal}_0(V, \mathcal{M})$ and $\mathrm{Gal}^+(V, \mathcal{M})$, or for a category $\mathrm{Gal}_0(V_P, \mathcal{M}_P)$ or $\mathrm{Gal}^+(V_P, \mathcal{M}_P)$. By abuse of language, we shall often call the arrows of Gal^+ galleries.

(1.26) Lemma.

(i) In Gal^+, for every gallery g, one has

$$g\Delta = \Delta(-g).$$

(ii) For any g and h with source A in Gal^+, there exists n such that $g\Delta^n$ begins with h. If h is composed of k galleries $u(A_i, A_{i+1})$, one may take $n = k$.

By induction, it is enough to prove (i) when g has length one. For $g = (B, C)$, one has

$$g\Delta = u(B, C)u(C, -C) = u(B, C)u(C, -B)u(-B, -C) = u(B, -B)u(-B, -C) = \Delta(-g).$$

For every chamber B, $\Delta = u(A, -A)$ begins with $u(A, B)$. Hence (ii) follows from (i) by induction on k.

The following proposition follows at once from (1.26), from its transform by $*$, and from (1.19)(i),(ii).

(1.27) Proposition.

(i) In Gal^+, the set of all arrows admits a left and right calculus of fractions.

(ii) Let $\mathrm{Gal}(V, \mathcal{M})$, or simply Gal, be the category, a groupoid, obtained from Gal^+ by making all arrows invertible. Call positive those arrows of Gal lying in the image of Gal^+. The canonical functor from Gal^+ to Gal is faithful, and every arrow g of Gal can be written in the forms

$$g = g_1 \Delta^{-n} = \Delta^{-n} g_2$$

with g_1 and g_2 positive.

If a category C has only one object A, and if the monoid $\mathrm{Hom}(A, A)$ is cancellative, to say that all arrows of C admit a left and right calculus of fractions means that $\mathrm{Hom}(A, A)$ satisfies Ore's condition on the left and on the right. The calculus of fractions used in (1.27) and below is an immediate generalization of Ore's theory for embedding such a monoid in a group.

(1.28) For every wall M, the number of times that $g \in \mathrm{Gal}^+$ crosses M is well-defined (1.11). This function of g extends additively to $g \in \mathrm{Gal}$. More generally, for every intersection of walls P, one could use the universal property of Gal to define functors

$$\pi_P : \mathrm{Gal}(V, \mathcal{M}) \longrightarrow \mathrm{Gal}(V_P, \mathcal{M}_P).$$

(1.29) Let A be a chamber, let P be an intersection of walls of A, and let $F = F(P)$ (1.6). The function π_F induces a functor

$$\pi_F : \mathrm{Gal}_0(V_P, \mathcal{M}_P) \longrightarrow \mathrm{Gal}_0(V, \mathcal{M}). \tag{1.29.1}$$

The image of this functor is stable under equivalence, and it induces a faithful functor

$$\pi_F : \mathrm{Gal}^+(V_P, \mathcal{M}_P) \longrightarrow \mathrm{Gal}^+(V, \mathcal{M}). \tag{1.29.2}$$

From the calculus of fractions and (1.19)(i) follows the following proposition.

(1.30) Proposition. Under the preceding hypotheses, the functor deduced from (1.29.2)

$$\pi_F : \mathrm{Gal}(V_P, \mathcal{M}_P) \longrightarrow \mathrm{Gal}(V, \mathcal{M})$$

is faithful.

(1.31) Proposition. Let C be a chamber; let I and J be two sets of walls of C; put $K = I \cup J$; let P, Q, R be the intersections of the walls in I, J, K, and let $F(P), F(Q), F(R)$ be the corresponding facets in C (1.6). The facet $F(R)$ is the smallest facet containing $F(P)$ and $F(Q)$. In the groupoid $\mathrm{Gal}(V, \mathcal{M})$, one has

$$\pi_{F(P)}(\mathrm{Gal}(V_P, \mathcal{M}_P)) \cap \pi_{F(Q)}(\mathrm{Gal}(V_Q, \mathcal{M}_Q)) = \pi_{F(R)}(\mathrm{Gal}(V_R, \mathcal{M}_R)).$$

A chamber having $F(P)$ and $F(Q)$ as facets also has $F(R)$ as a facet. This proves (1.31) for objects.

Let $g \in \mathrm{Hom}_{\mathrm{Gal}}(A, B)$, and suppose that

$$g = \pi_{F(P)}(g_1) = \pi_{F(Q)}(g_2).$$

By (1.27)(ii), one has

$$g_1 = \Delta(P)^{-n} g_1', \qquad g_2 = \Delta(Q)^{-n} g_2',$$

with g_i' positive, for $n > 0$ sufficiently large. Let $g_1 = \pi_{F(P)}(g_1')$ and $g_2 = \pi_{F(Q)}(g_2')$. By (1.26)(ii), transformed by $*$, $\Delta^n \Delta(P)^{-n}$ is positive, and in $\mathrm{Gal}^+(V, \mathcal{M})$ one has

$$(\Delta^n \Delta(P)^{-n}) g_1 = (\Delta^n \Delta(Q)^{-n}) g_2. \tag{1.31.1}$$

(1.31.2) Lemma. If $h \in \mathrm{Hom}_{\mathrm{Gal}^+}(A, B)$ begins both with $\Delta^n\Delta(P)^{-n}$ and with $\Delta^n\Delta(Q)^{-n}$, then h begins with $\Delta^n\Delta(R)^{-n}$.

We deduce (1.31) from (1.31.2). By (1.31.2), one has

$$(\Delta^n\Delta(P)^{-n})g_1 = (\Delta^n\Delta(Q)^{-n})g_2 = (\Delta^n\Delta(R)^{-n})g_3$$

with g_3 positive. Hence $g = \Delta(R)^{-n}g_3$. The positive gallery $g_3 = \Delta(R)^n g$ lies in the image of both $\pi_{F(P)}$ and $\pi_{F(Q)}$. By (1.28), it therefore crosses zero times every wall not containing P or Q, i.e. every wall not containing R. It thus lies in the image of $\pi_{F(R)}$.

We prove (1.31.2). Let $A(P) = A.\Delta.A(P)$, and similarly for Q and R. The gallery $\Delta\Delta(P)^{-1}$ with source A is $u(A, A(P))$; the one with source $A(P)$ is $u(A(P), A)$. By (1.26)(i), $\Delta\Delta(P) = \Delta(P)\Delta$. Hence

$$\Delta^n\Delta(P)^{-n} = u(A, A(P))\, u(A(P), A)\, u(A, A(P)) \cdots \tag{1.31.3}$$

with n factors.

(1.31.4) Lemma. $A(R) \in D(A(P), A(Q))$.

With the notation of (1.2), $\mathcal{M}(A, A(P))$ is the set of walls containing P, and $\mathcal{M}(A, A(Q))$ is the set of those containing Q. By (1.2), $\mathcal{M}(A(P), A(Q))$ is the set of walls containing P or Q, but not R, namely

$$\mathcal{M}(A(P), A(R)) \cup \mathcal{M}(A(Q), A(R)),$$

and one applies (1.4).

By (1.19)(iii), a gallery class which begins with $\Delta\Delta(P)^{-1}$ and $\Delta\Delta(Q)^{-1}$ also begins with $\Delta\Delta(R)^{-1}$, and (1.31.2) follows by induction from the following lemma.

(1.31.5) Lemma. Let P and R be intersections of walls of a chamber A, with $P \subset R$. Let h be in Gal^+ with source A. If h begins both with $\Delta^n\Delta(P)^{-n}$, $n > 0$, and with $\Delta\Delta(R)^{-1}$, then h begins with

$$(\Delta\Delta(R)^{-1})(\Delta\Delta(P)^{-1})^{n-1}.$$

One may assume $n > 2$. Put $h = (\Delta\Delta(P)^{-1})h'$. With the preceding notation, h' then begins both with $\Delta\Delta(P)^{-1} = u(A(P), A)$ and with $u(A(P), A(R))$. Apply (1.19)(iii). The chamber C of loc. cit. is then separated from $A(P)$ by every wall M separating $A(P)$ from A, i.e. $M \not\supset P$, or $A(P)$ from $A(R)$, i.e. $M \supset P$ and $M \not\supset R$. Thus C is separated from $A(P)A$ only by walls $M \supset R$, and

$$C \in D(A(P)A, A(P)A(R)).$$

It follows that h' begins both with $\Delta\Delta(R)^{-1}$ and with $(\Delta\Delta(P)^{-1})^{n-1}$. Induction on n shows that h' begins with

$$(\Delta\Delta(R)^{-1})(\Delta\Delta(P)^{-1})^{n-2},$$

so that h begins with

$$(\Delta\Delta(P)^{-1})(\Delta\Delta(R)^{-1})(\Delta\Delta(P)^{-1})^{n-2}.$$

One concludes by noting that

$$\Delta\Delta(P)^{-1}\Delta\Delta(R)^{-1} = \Delta\Delta(R)^{-1}\Delta\Delta(P)^{-1}.$$

(1.32) Proposition. For F a facet of a chamber C, generating an intersection of walls P, one has

$$\pi_F^{-1}(\mathrm{Gal}^+(V, \mathcal{M})) = \mathrm{Gal}^+(V_P, \mathcal{M}_P) \subset \mathrm{Gal}(V_P, \mathcal{M}_P).$$

Suppose that $\pi_F(\Delta^{-n}g) = h$ lies in $\mathrm{Gal}^+(V, \mathcal{M})$. Then $\pi_F(g) = \Delta(P)^n h$, and by (1.28) every gallery H representing h crosses only walls passing through F. Thus $h = \pi_F(h_1)$, with h_1 in $\mathrm{Gal}^+(V_P, \mathcal{M}_P)$, and the conclusion follows from (1.30).

2. Buildings

(2.1) Let $(V, \mathcal{M})$ be as in §1, satisfying (1.8), and let $r = \dim V$. Let S be the sphere of radius one in V, for some Euclidean structure on V; more intrinsically, one could take $S = (V - \{0\})/\mathbb{R}_+^*$. The hyperplanes $M \in \mathcal{M}$ cut out a triangulation of S, and we shall again denote by S both the corresponding simplicial space and its geometric realization. We transfer to S the terminology used for V: chambers, facets, adjacency, galleries.

(2.2) Choose a fundamental chamber A_0 in S. We denote by I^+ the space constructed below; it depends on $V, \mathcal{M}, A_0$, and is equipped with $q : I^+ \to S$.

a) Let $\mathcal{G}^+$ be the set of equivalence classes of galleries g with source A_0,

$$\mathcal{G}^+ = \coprod_B \mathrm{Hom}_{\mathrm{Gal}^+}(A_0, B).$$

b) Let Z^+ be the disjoint sum, indexed by $g \in \mathcal{G}^+$, of the closed simplex of S which is the closure of the open simplex of dimension $r-1$ equal to the target of g:

$$Z^+ = \coprod_{g \in \mathcal{G}^+} (\mathrm{target}\, g)^-.$$

Let q' be the evident map from Z^+ to S.

c) I^+, with q, is a quotient of Z^+, with q'. If G is a gallery from A_0 to B, and if C is a chamber adjacent to B, one glues $(\mathrm{target}\, g)^-$ and $(\mathrm{target}\, g(BC))^-$ along the common closed face of their images in S.

(2.3) The space I^+ is decomposed into facets, images of facets of Z^+, each mapping bijectively to a facet of S. Its chambers, respectively faces and vertices, are its facets of dimension $r-1$, respectively $r-2$ and 0.

The vertices of a facet F are the vertices contained in the closure $\overline{F}$ of F. The chambers of I^+ are indexed by $\mathcal{G}^+$; $A_0.g$ denotes the one indexed by g, and A_0 the one indexed by (A_0).

Let $g \in \mathrm{Hom}_{\mathrm{Gal}^+}(A_0, A)$, let $A = A_0.g$, and let $H = (C_0, \ldots, C_n)$ be a gallery from A to B, with class h. Put

$$A.h = A_0.gh.$$

The chambers $A.(C_0, \ldots, C_i)$, $0 \le i \le n$, form a gallery in I^+. Such galleries will be called positive.

(2.4) Lemma. Let F be a facet of S, let A and B be two chambers of S having F as a facet, and let $g \in \mathrm{Hom}_{\mathrm{Gal}^+}(A_0, A)$, $h \in \mathrm{Hom}_{\mathrm{Gal}^+}(A_0, B)$. The following conditions are equivalent:

(a) in I^+, the facets $q^{-1}(F)$ of $A_0.g$ and $A_0.h$ coincide;

(b) $g^{-1}h \in \mathrm{Hom}_{\mathrm{Gal}}(A, B)$ lies in $\pi_F(\mathrm{Gal})$;

(c) there are g_1 and h_1 in $\pi_F(\mathrm{Gal}^+)$ such that $gg_1 = hh_1$.

Condition (b) is an equivalence relation between galleries from A_0 to a chamber having F as a facet. This gives (a) $\Rightarrow$ (b). That (b) implies (c) follows from (1.26)(i). Finally, (c) $\Rightarrow$ (a) is immediate from the definitions.

From (2.4), (a) $\Leftrightarrow$ (b), and from (1.31), one immediately obtains:

(2.5) Proposition. Each facet of I^+ is uniquely determined by the set of its vertices. Thus I^+ may be described as the geometric realization of the following simplicial scheme.

a) For x a vertex of S, let $\mathcal{G}(x)$ be the set of g in Gal^+ with source A_0 and with target a chamber having x as a vertex. Let $\mathcal{G}(x)/\pi_x$ be the quotient of $\mathcal{G}(x)$ by the equivalence relation (2.4)(b), for $F = x$. Then the set of vertices is

$$\coprod_x \mathcal{G}(x)/\pi_x.$$

b) A set E of vertices spans a simplex if and only if there exists a gallery g with source A_0 such that, for $y \in E$, g lies in $\mathcal{G}(qy)$ and y is the image of g in $\mathcal{G}(qy)/\pi_{qy}$.

(2.6) Proposition. Let F be a facet of I^+. There exists a gallery class g such that, for F to be a facet of a chamber $B = A_0.h$, it is necessary and sufficient that

$$h = g\,\pi_F(h')$$

for a suitable h' in Gal^+.

Take g to be a gallery of minimum length such that F is a facet of $A = A_0.g$. Let $B = A_0.h$ be a chamber having F as a facet. By (2.4)(a) $\Leftrightarrow$ (c), there are galleries h_1 and h_2 in $(V_{qF}, \mathcal{M}_{qF})$ with

$$g\pi_F(h_1) = h\pi_F(h_2).$$

Apply the transform by $*$ of (1.24). By the minimality of g, one finds that h_1 ends with h_2, say $h_1 = h'h_2$. By (1.19)(ii), $g\pi_F(h') = h$, which proves (2.6).

(2.7) Notation. If A is a chamber of I^+, denote by $S(A)$ the union of the closed chambers $(A.(qA, B))^-$, where B runs through the chambers of S. One checks that $q|S(A)$ is an isomorphism from $S(A)$ to S.

(2.8) Definition. $\overline{I}^+$ is the space obtained from I^+ by filling in the spheres $S(A)$.

More precisely, let B^r be a ball whose boundary is S. If one wants to work simplicially, one takes B^r to be the cone on the simplicial space S. Define $\overline{I}^+$ from I^+ by attaching a family of copies $b(A)$ of B^r, indexed by the chambers of I^+.

(2.9.2) Lemma. Suppose that there exists a chamber $B = A_0.h$ in $(I^+)_{n+1}$, distinct from $A = A_0.g$, having F as a facet. Then there exists a chamber $B' = A_0.h'$ in $(I^+)_n$, and a wall M of qB', containing qF, such that

$$A = B'.\Delta(M).$$

Let g_0 be the gallery considered in (2.6). One has

$$g = g_0\pi_F(g_1), \qquad h = g_0\pi_F(h_1).$$

The hypotheses imply that $\mathrm{lg}(g_1) \neq 0$. For a suitable wall M of qA, one then has $g_1 = h'\Delta(M)$, and one puts $B' = \pi_F h'$.

For the closed chamber $\overline{A}$, $\overline{A} \cap (I^+)_n$ is the union of a nonempty set of closed faces, and in $(I^+)_{n+1}$ the points of

$$\overline{A} - (\overline{A} \cap (I^+)_n)$$

belong only to the single closed chamber $\overline{A}$. There are two cases.

Case 1. $\overline{A} \cap (I^+)_n \neq \partial\overline{A}$. In this case, there is no sphere $S(B)$ with

$$A \subset S(B) \subset (I^+)_{n+1}.$$

We take $\varphi_t|\overline{A}$ to be a collapse of $\overline{A}$ onto $\overline{A} \cap (I^+)_n$.

Case 2. $\overline{A} \cap (I^+)_n = \partial\overline{A}$. Put $A = A_0.g$. For every wall M of qA, it follows from (2.7.2) that g ends with $\Delta(M)$. By (1.19)(iii), g therefore ends with Δ: $A = B.\Delta$, with B uniquely determined by A, by (1.19)(ii). In passing from $(\overline{I}^+)_{n+1}$ to $(\overline{I}^+)_n$, A and the interior of the ball $b(B)$ disappear. We take $\varphi_t|b(B)$ to be a collapse of $b(B)$ onto $S(B) - A$. As seen in case 1, every ball which disappears is of the preceding type. This completes the construction of φ_t, and proves (2.7).

(2.10) The space I, with $q : I \to S$, is defined like I^+, replacing Gal^+ by Gal:

a) put $\mathcal{G} = \coprod_B \mathrm{Hom}_{\mathrm{Gal}}(A_0, B)$;

b) put $Z = \coprod_{g\in\mathcal{G}} (\mathrm{target}\, g)^-$;

c) let q' be the evident map from Z to S. The space I, with q, is a quotient of Z, with q'. If $g \in \mathcal{G}$ has target B and C is adjacent to B, one glues $(\mathrm{target}\, g)^-$ and $(\mathrm{target}\, g(BC))^-$ along the common closed face of their images in S.

Equivalently, if two chambers B and C have a common facet F, if $g \in \mathrm{Hom}_{\mathrm{Gal}}(A_0, B)$, and if $h \in \mathrm{Hom}_{\mathrm{Gal}}(A_0, C)$, then, in I, the closed facets $q'^{-1}(F)^-$ of $(\mathrm{target}\, g)^-$ and of $(\mathrm{target}\, h)^-$ are identified if and only if

$$hg^{-1} \in \pi_F\, \mathrm{Hom}_{\mathrm{Gal}}(\pi_F^{-1}B, \pi_F^{-1}C).$$

(2.11) As for I^+, one defines chambers, faces, and open or closed facets of I. One defines A_0, and $A.h$, for A a chamber and h in Gal with source qA, as in (2.3). The analogue of (2.4)(a) $\Leftrightarrow$ (b) is here trivially true. As in (2.5), one deduces from (1.31) that each facet of I is uniquely determined by its set of vertices, and I appears as the geometric realization of a simplicial scheme with a description of type (2.5). For every chamber A of I, define a sphere $S(A)$ as in (2.7); q induces an isomorphism from $S(A)$ to S. As in (2.8), define a building $\overline{I}$, equipped with $q : \overline{I} \to B^r$, by filling in all the spheres $S(A)$.

(2.12) Besides q and its chamber decomposition, I has the following additional structure. A composition $B.g$ is defined, which associates to a chamber B of I, whose image in S is B, and to $g \in \mathrm{Hom}_{\mathrm{Gal}}(B, C)$, a chamber of I whose image in S is C. The construction $g \mapsto B.g$ identifies the analogue of I obtained by choosing B as fundamental chamber with the space I. In particular,

$$B.(gh) = (B.g).h.$$

(2.13) For every chamber A of I above A_0, define a map

$$i_A : I^+ \longrightarrow I$$

and similarly $i_A : \overline{I}^+ \to \overline{I}$, compatible with the projection to S, respectively B^r, by sending the closed chamber $(A.g)^-$ of I^+, respectively the ball $b(A.g)$ of $\overline{I}^+$, to the chamber, respectively ball, of the same name in I, respectively $\overline{I}$.

(2.14) Proposition.

(i) i_A identifies I^+ with a closed subspace of I, and $\overline{I}^+$ with a closed subspace of $\overline{I}$.

(ii) One has

$$I = \varinjlim i_{A_0.\Delta^{-2n}}(I^+), \qquad \overline{I} = \varinjlim i_{A_0.\Delta^{-2n}}(\overline{I}^+).$$

The assertions (i) or (ii) for $\overline{I}^+$ and $\overline{I}$ follow from the corresponding assertions for I^+ and I.

We prove (i) for I^+ and I. Let B and C be two chambers of S with a common facet F. Let $g \in \mathrm{Hom}_{\mathrm{Gal}^+}(A_0, B)$ and $h \in \mathrm{Hom}_{\mathrm{Gal}^+}(A, C)$. Suppose that the facets $q^{-1}F$ of $(A.g)^-$ and $(A.h)^-$ are equal in I. Then

$$g^{-1}h = \pi_F(e) \quad \text{for} \quad e \in \mathrm{Hom}_{\mathrm{Gal}}(\pi_F^{-1}B, \pi_F^{-1}C).$$

By (1.26), write $e = e_1 e_2^{-1}$, with e_i in Gal^+. By (1.29), $g\pi_F(e_1) = h\pi_F(e_2)$ in Gal^+, and the facets $q^{-1}F$ of $(A.g)^-$ and $(A.h)^-$ are therefore already equal in I^+.

For (ii), it is enough to note that every gallery g with source A_0 in Gal can be written

$$g = \Delta^{-2n} g'$$

with g' in Gal^+.

From (2.14) one obtains $\pi_i(I) = \varinjlim \pi_i(I^+)$. Since I is a CW-complex, (2.9) gives:

(2.15) Theorem. The space $\overline{I}$ is contractible.

The space I is therefore the bouquet of the spheres $S(A)$, as A ranges through the chambers of I.

3. Coverings

(3.1) Let $V_{\mathbb{C}}$ be the complexification of V, and let $M_{\mathbb{C}}$ be the complexification of M, for $M \in \mathcal{M}$. Put

$$Y = V_{\mathbb{C}} - \bigcup_{M \in \mathcal{M}} M_{\mathbb{C}}.$$

In this section, we construct by gluing a space above Y. The gluing data will be described using I and $\overline{I}$. The facets of S and the facets of V other than $\{0\}$ are in canonical bijection, and we shall constantly pass from one to the other. In general we shall use the same notation for the corresponding facets; when necessary the correspondence will be denoted $F \mapsto [F]$.

(3.2) Lemma. Let A, B, C be three chambers of I, with $C \subset S(A) \cap S(B)$. Then q induces an isomorphism from $S(A) \cap S(B)$ to the intersection of S with the closed half-spaces $D'_M(qC)$, containing qC, bounded by a wall M which separates qA from qB.

The intersection of the $D'_M(qC)$ is a union of closed chambers. If C' is one of them, then qA and qB lie on the same side of every wall separating qC from C'. It follows that the lifts, to $S(A)$ or $S(B)$, of a minimal gallery from qC to C' coincide, and $(C')^- \subset q(S(A) \cap S(B))$.

Conversely, if a facet F is not in the intersection of the $D'_M(qC)$, then there is a wall M such that $F \subset M$, separating qA from qB and F from qC. Suppose, for contradiction, that $F \subset q(S(A) \cap S(B))$. Let D be a chamber of S having F as a facet, and let $(C_0, C_1, \ldots, C_{n+1})$ be a gallery from qC to D. Let M_i be the wall separating C_i from C_{i+1}, and let α_i, respectively β_i, be $+1$ or -1 according as C_{i+1} and A, respectively B, are or are not on the same side of M_i. By hypothesis, F is a facet of

$$C.\Delta(M_0)^{\alpha_0} \cdots \Delta(M_n)^{\alpha_n}$$

and of

$$C.\Delta(M_0)^{\beta_0} \cdots \Delta(M_n)^{\beta_n}.$$

By (2.4) and (1.24), since $M \not\supset F$, this gives contradictory equalities: one side equals 1, the other -1.

(3.3) Lemma. Let $\Re$ and $\Im$ be the real and imaginary part maps from $V_{\mathbb{C}}$ to V. Let $v \in V_{\mathbb{C}}$, and let F be the facet of V such that $\Re v \in F$. For v to belong to Y, it is necessary and sufficient that $\mathrm{pr}_F(\Im v)$ lie in a chamber of $(V_F, \mathcal{M}_F)$.

This is clear.

What follows is based on the following intuition: to every walk in I, performed by following a gallery of the form

$$\Delta(M_1)^{\varepsilon_1} \cdots \Delta(M_k)^{\varepsilon_k}, \qquad \varepsilon_i = \pm 1,$$

there corresponds a path, or path class, in Y. Its image under $\Re$ in V passes from chamber to chamber, crossing successively the walls $M_1, \ldots, M_k$ at points m_i. For $\varepsilon_i = 1$, respectively -1, the path passes through one of the two components R of $\Re^{-1}(m_i) \cap Y$, in bijection by $\Im$ with $V - M_i$: the one such that $\Im R$ contains, respectively does not contain, the preceding chamber.

(3.4) Notation.

(i) If C is a chamber of $(V, \mathcal{M})$, let $Y(C)$ be the open subset of Y consisting of the $x \in V_{\mathbb{C}}$ satisfying the following condition: if F is the facet of V such that $\Re x \in F$, then $\mathrm{pr}_F(\Im x) \in \pi_F^{-1}(C)$.

(ii) Let A and B be two chambers of I. If $S(A) \cap S(B)$ contains a chamber C, temporarily denote by $Y'(A, B)$ the intersection of the open half-spaces $D'_M(qC)^\circ$ for M as in 3.2. Put

$$Y(A, B) = Y(qA) \cap Y(qB) \cap \Re^{-1}(Y'(A, B)) \subset V_{\mathbb{C}}.$$

If $S(A) \cap S(B)$ contains no chamber, put $Y(A, B) = \varnothing$.

(iii) For a facet F of V, the star $\mathrm{Et}(F)$ is the union of the open facets G of $(V, \mathcal{M})$ such that $F \subset \overline{G}$.

(iv) For a facet F of V and a chamber C of $(V_F, \mathcal{M}_F)$, put

$$V(F, C) = \{\, v \in V_{\mathbb{C}} \mid \Re v \in \mathrm{Et}(F),\ \mathrm{pr}_F(\Im v) \in C \,\}.$$

(3.5) Lemma. Let G be a facet of V. The $V(F, C)$, for F a facet such that $G \subset \overline{F}$ and C a chamber of V_F, form an open covering of

$$Y \cap \Re^{-1}(\mathrm{Et}\, G).$$

In particular, taking $G = \{0\}$, the $V(F, C)$ cover Y.

If $x \in Y \cap \Re^{-1}(\mathrm{Et}\, G)$ and $\Re x \in F$, then x lies in one of the $V(F, C)$, by (3.3).

(3.6) Let Y' be the disjoint sum, indexed by the balls $b(A)$ of $\overline{I}$,

$$Y' = \coprod_{b(A)} Y(qA).$$